VANSHU GALHOTRA

Master of Computer Applications
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Educational Qualification

Year	Degree/Examination	Institution/Board	CGPA/Percentage
2027	M.C.A	NIT, Trichy	09.55*
2024	B.C.A	Panjab University, Chandigarh	85.91%
2021	Class XII	DAV Public School, Kotkapura, CBSE	98.20%
2019	Class X	DAV Public School, Kotkapura, CBSE	90.60%

Academic Achievements

- BCA **(First Position)**: DAV College, Chandigarh (Panjab University)
- **1st Prize**: Software Development Competition Inter College (Panjab University)
- **2nd Prize**: Debugging Competition Inter College (Panjab University)
- **2nd Prize**: Mathematics Quiz Intra College

Internship Experience

- **Research Internship at DCSA, Panjab University, Chandigarh**: *May 2025 – Jun 2025*
  Worked on the project **"Handwritten Trajectories Identification in Scanned Text"**, focused on recovering stroke order from static handwritten images. The project involved implementing **graph-theoretic and heuristic-based methods** to extract plausible pen trajectories. A neural network was used to recognize the recovered feature vector, achieving a recognition accuracy of **98.90%**. The results were at par with existing literature. The work also led to the preparation of a **research paper** during the internship.

Other Projects

- **Real time Fleet Management System:** *Oct 2025 – Jan 2026*
  Designed and developed a production-grade real-time fleet management system for a private logistics company to manage vehicle records, document tracking, and operational workflows. Built using Next.js for the frontend and NestJS for the backend with a PostgreSQL database. Deployed on AWS with Nginx as a reverse proxy, and implemented automated S3 backups for data reliability. The system is currently live and actively used in daily operations.

- **BackTrack – Reverse Engineering Event Platform (INFOTREK'25):** *May 2025 – Jun 2025*
  Developed **BackTrack**, a web platform created for the INFOTREK'25 event organised by **ACM**. The website is specifically used for managing and hosting the reverse engineering event. Built using **Next.js** for the frontend and **NestJS** for the backend, the project also incorporates **CI/CD pipelines** for automated deployment and smooth development workflow.

- **Python Chess Engine Library (pychess-engine):** *Sep 2024 – Dec 2024*
  Developed **pychess-engine**, a Python based chess engine library available on **PyPI.** It is designed to evaluate board positions, calculate the best moves, and support integration with various interfaces. The engine follows object-oriented principles for a modular structure, uses alpha-beta pruning for efficient move evaluation, and applies multi-threading to improve performance.

Areas of Interest

- Data Structures and Algorithms
- Design and Analysis of Algorithms
- Machine Learning
- Deep Learning

Technical Skills and Certifications

- Programming Languages : C, C++, Python, JavaScript (Basic), Java (Basic)
- Web Development : HTML, CSS, JavaScript, React.js, Node.js, Next.js, SQL, MongoDB, AWS
- Machine Learning : Python, NumPy, Pandas, Scikit-learn, MLP, CNN
- Other Software : Microsoft Word, GitHub, Visual Studio Code, LaTeX
- Certifications : Crash Course on Python (Coursera offered by Google)

Positions of Responsibility

- **Chairperson – ACM Student Chapter NIT Trichy:** *Oct 2025 - Present*
  Responsible for helping organize technical events and workshops under the student chapter. Contributed to the planning of the INFOTREK and other activities aimed at building interest in tech among students.
- **EEC Head, VERSION'26** *Nov 2025 - Present*
  Managed team to plan, develop, and execute events for VERSION'26.

Extracurricular Activities

Sport Activities:

- Basketball – **Sportsfete'24**, the inter department sports fest organized by NIT Trichy. Supported the department team during the event.